Ch 10.2: Multi-Layer Neural Nets

Lecture 30 - CMSE 381

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Announcements

Last time:

- Single Layer Neural Nets

This lecture:

- Multi-layer Neural Nets
- Application to MNIST

Announcements:

- HW 9 Due Sunday 11/24
- Exam 3 12/4
- Project Due 12/6

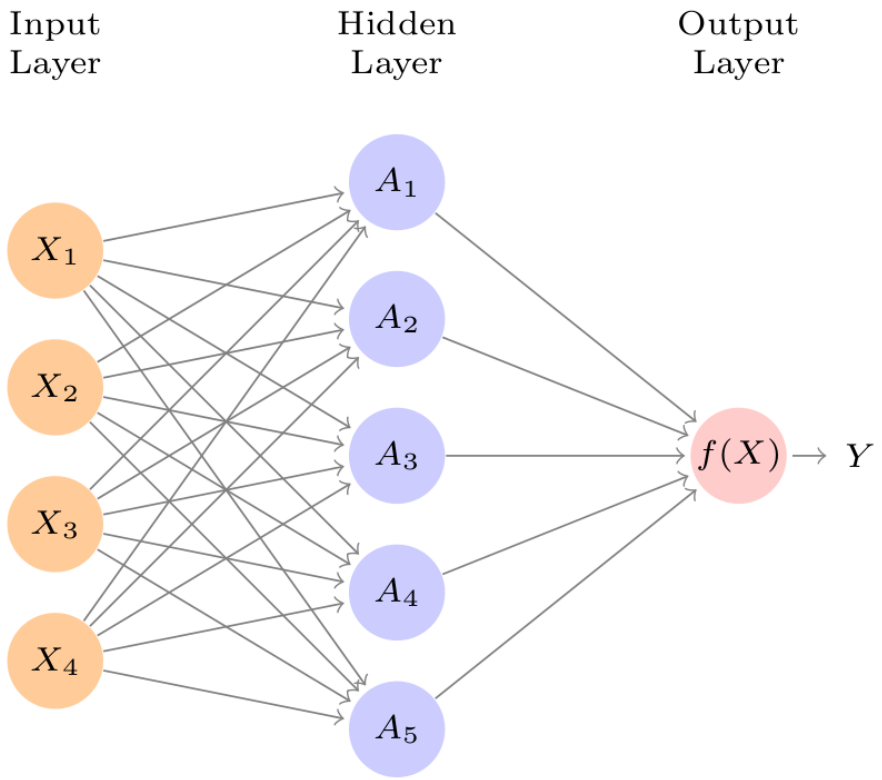
The end is near!

Lec #	Date			Reading	HW
31	Wed	11/20	Multi Layer NN	10.2	
32	Fri	11/22	CNN	10.3	HW #11 Due Sun 11/24
33	Mon	11/25	TBD: Unsupervised learning/clustering	12.1, 12.4?	
	Wed	11/27	Virtual: Project office hours		
	Fri	11/29	No class - Thanksgiving		
	Mon	12/2	Review		
	Wed	12/4	**Midterm #3**		
	Fri	12/6	No class - EGR Design Day		Project due

Section 1

Neural Nets

Feed Forward Neural Network: The cartoon



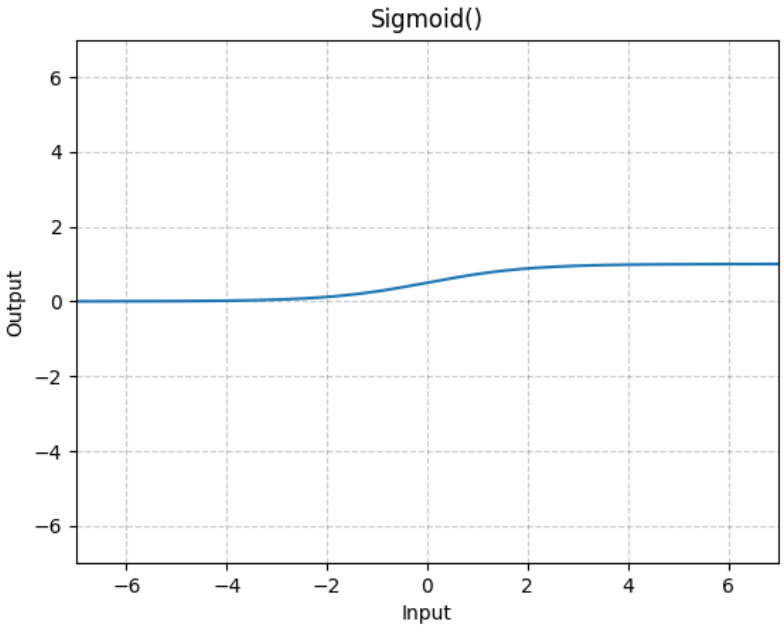
$$A_k = h_k(X) = g(w_{k0} + \textstyle\sum_{j=1}^{p} w_{kj} X_j),$$

$$f(X) = \beta_0 + \sum_{k=1}^{K} \beta_k A_k$$

Choices for activation function

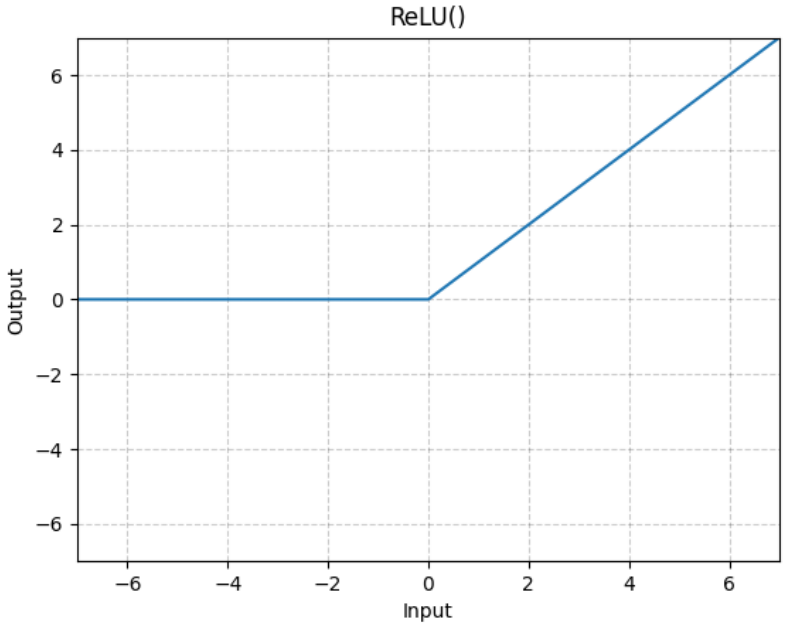
Sigmoid:

$$g(z) = \frac{e^z}{1+e^z} = \frac{1}{1+e^{-z}}$$

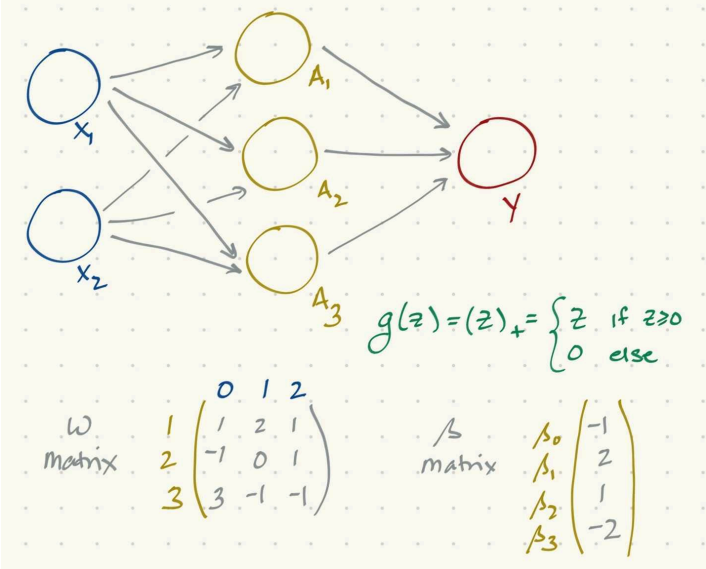


ReLU: Rectified linear unit

$$g(z) = (z)_+ = \begin{cases} 0 & \text{if } z < 0 \\ z & \text{else.} \end{cases}$$



Matrix version



$$A_k = h_k(X) = g(w_{k0} + \textstyle\sum_{j=1}^{p} w_{kj} X_j),$$

$$A = g(\mathbf{W} \cdot \mathbf{X}) \qquad \mathbf{X}^T = (1\ X_1\ X_2\ \cdots\ X_p)$$

$$f(X) = \beta_0 + \sum_{k=1}^{K} \beta_k A_k$$

$$Y = \beta \cdot \mathbf{A} \qquad \mathbf{A}^T = (1\ A_1\ A_2\ \cdots\ A_K)$$

Training the model

Choose parameters by minimizing RSS, $\sum_{i=1}^{n}(y_i - f(x_i))^2$ (or other loss function)

Chosen in advance:

The Architecture

- # of hidden layers
- # of hidden units
- activation function g

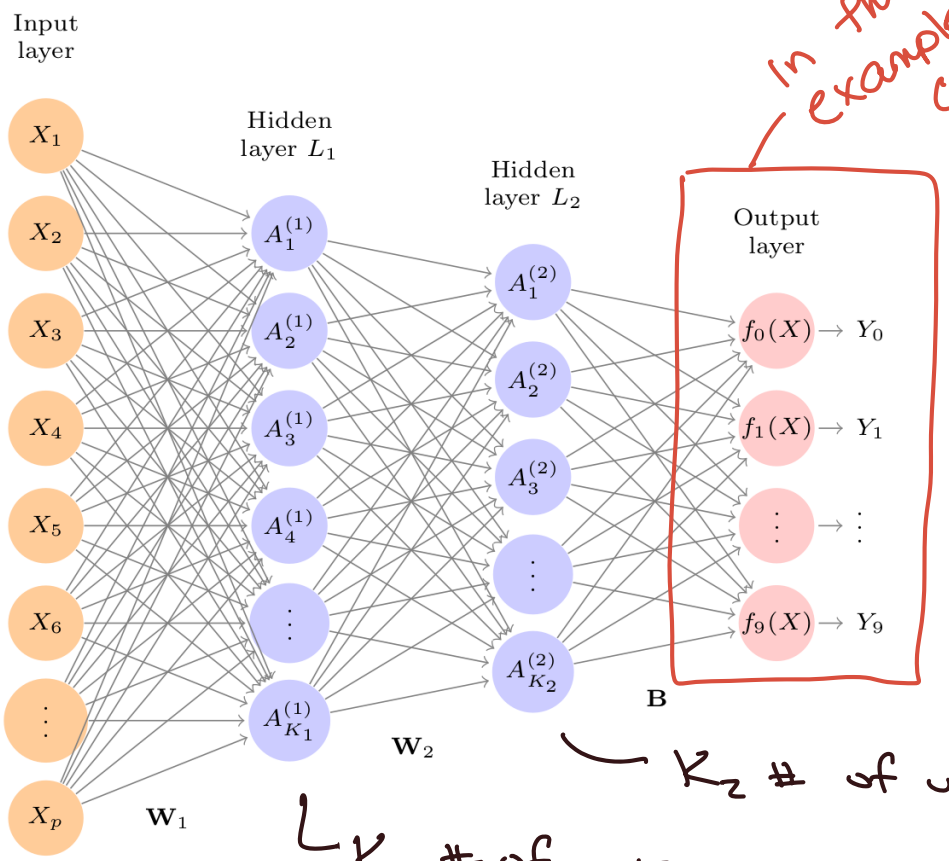
Tuned by the model:

- w
- β

Section 2

Multilayer Neural Networks

Multiple layers



K_1 # of units

K_2 # of units

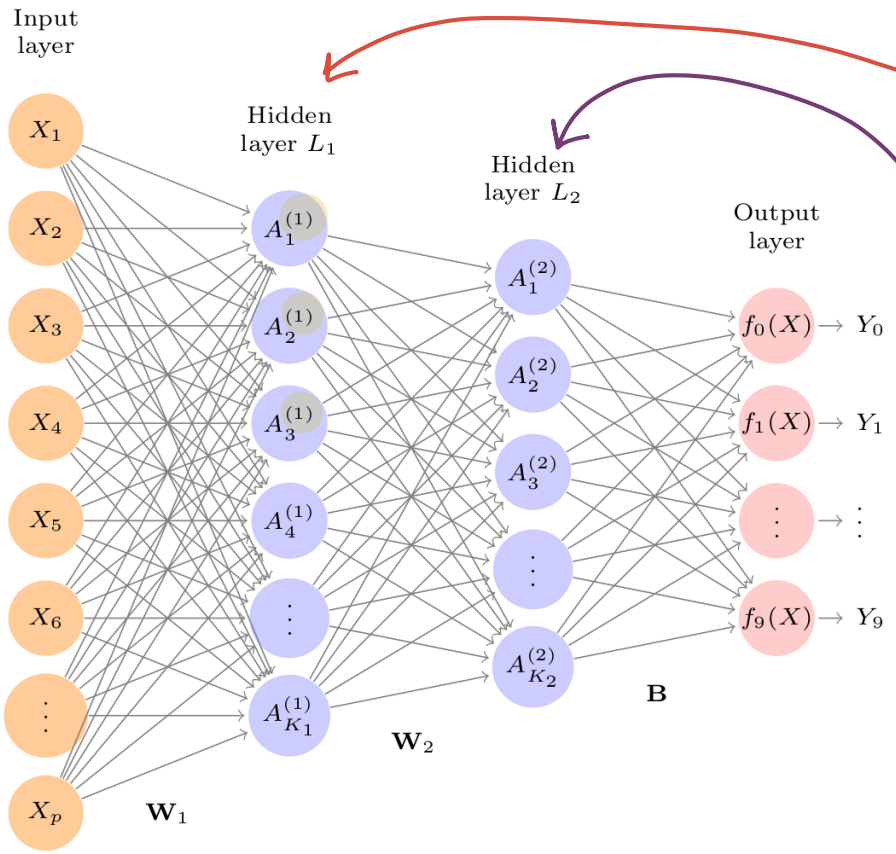
In this example ... classification more on that later.

Can include more than one internal lay

Can decide how many units per lay

each layer is linear combo of previous w/ activation function.

Hidden layers

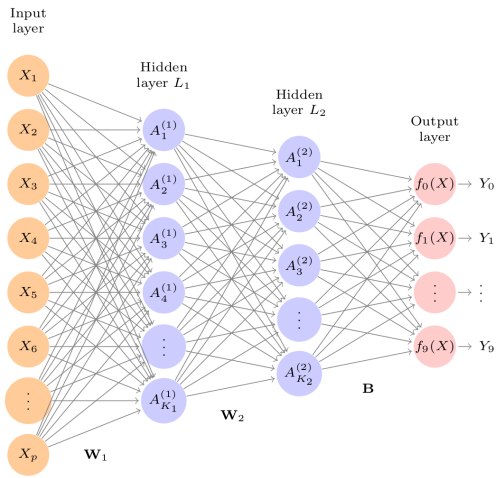


$$A_k^{(1)} = h_k^{(1)}(X) = g(w_{k0}^{(1)} + \sum_{j=1}^{p} w_{kj}^{(1)} X_j)$$

$$A_\ell^{(2)} = h_\ell^{(2)}(X) = g(w_{\ell 0}^{(2)} + \sum_{k=1}^{K_1} w_{\ell k}^{(2)} A_k^{(1)})$$

ex) 2 hidden layers
L_1 w/ 256 units
L_2 w/ 128 units
10 output layer → classification w/ 10 classes

More on that architecture



$$
\begin{aligned}
A_k^{(1)} &= h_k^{(1)}(X) \\
&= g(w_{k0}^{(1)} + \textstyle\sum_{j=1}^{p} w_{kj}^{(1)} X_j) \\
A_\ell^{(2)} &= h_\ell^{(2)}(X) \\
&= g(w_{\ell 0}^{(2)} + \textstyle\sum_{k=1}^{K_1} w_{\ell k}^{(2)} A_k^{(1)})
\end{aligned}
$$

W_1 entire matrix of weights

$\Rightarrow 785 \times 256 \cong 200k$ values

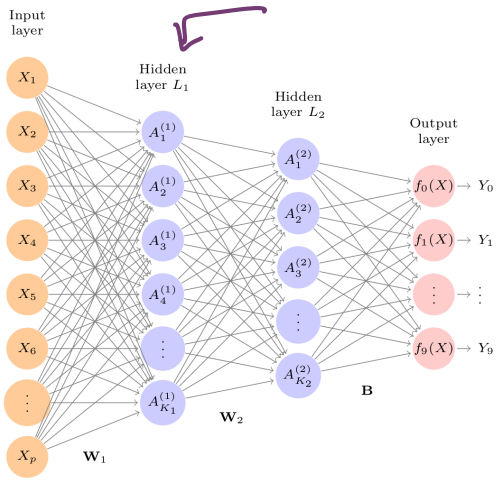
ex) 2 hidden layers

L_1 w/ 256 units

L_2 w/ 128 units

10 output layer → classificator w/ 10 classes

Matrix version: First layer

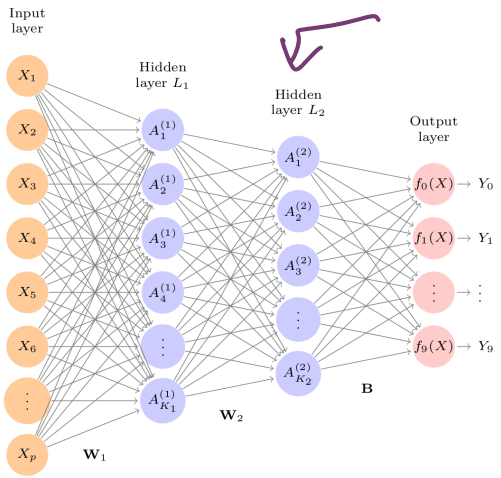


$$
\begin{aligned}
A_k^{(1)} &= h_k^{(1)}(X) \\
&= g(w_{k0}^{(1)} + \textstyle\sum_{j=1}^{p} w_{kj}^{(1)} X_j)
\end{aligned}
$$

$$\underline{\underline{A^{(1)}}} = g(\mathbf{W}^{(1)} \cdot \mathbf{X}) \qquad \mathbf{X}^T = (1\ X_1\ X_2\ \cdots\ X_p)$$

(same as for single layer with extra super script)

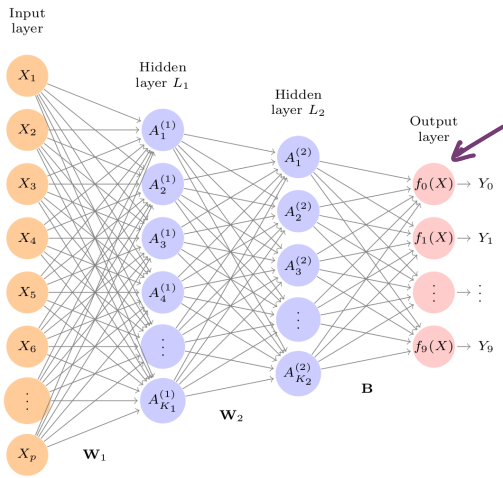
Matrix version: Second layer



$$
\begin{aligned}
A_\ell^{(2)} &= h_\ell^{(2)}(X) \\
&= g(w_{\ell 0}^{(2)} + \textstyle\sum_{k=1}^{K_1} w_{\ell k}^{(2)} A_k^{(1)})
\end{aligned}
$$

$$
A^{(2)} = g(\mathbf{W}^{(2)} \cdot \mathbf{A}^{(1)}) \qquad (\mathbf{A}^{(1)})^T = (1 \; A_1^{(1)} \; A_2^{(1)} \; \cdots \; A_{K_1}^{(1)})
$$

Matrix version: Last layer, first step



$$\boxed{Z_m} = \beta_{m0} + \sum_{\ell=1}^{K_2} \beta_{m\ell} h_\ell^{(2)}(X)$$

$$= \beta_{m0} + \sum_{\ell=1}^{K_2} \beta_{m\ell} A_\ell^{(2)},$$

$$\mathbf{Z} = \beta \cdot \mathbf{A}^{(2)}$$

matrix rather than vector

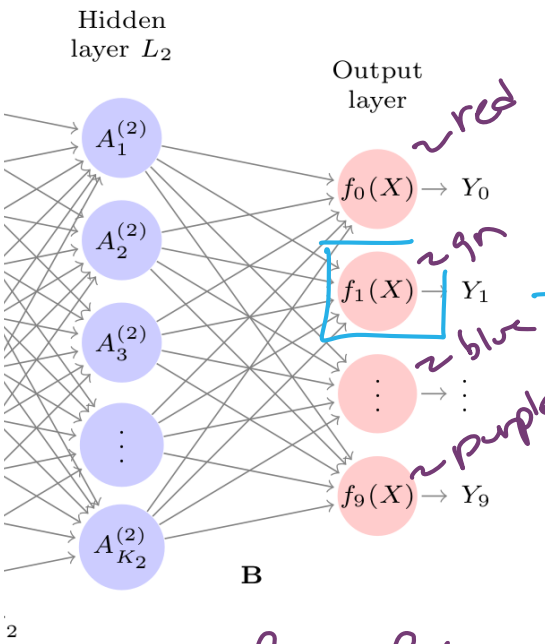
β is $M \times (K_2+1)$ matrix $\qquad (\mathbf{A}^{(2)})^T = (1\ A_1^{(2)}\ A_2^{(2)}\ \cdots\ A_{K_2}^{(2)})$

Goal: get these output functions into something that looks like a probability

The last column for classification: Softmax

$(z_0, \ldots, z_9)$ 10 different classes

$(e^{z_0}, \ldots, e^{z_n})$



$$f_m(X) = \Pr(Y = m|X) = \frac{e^{Z_m}}{\sum_{\ell=0}^{9} e^{Z_\ell}},$$

$$\frac{e^{z_{green}}}{\sum e^{z_i}}$$

- acts like probability
  - values sum to 1
  - values in [0, 1]

$f_0(x) + f_1(x) + f_2(x) + \ldots + f_9(x)$

- Returns class with highest value

An example

$$f_m(X) = \Pr(Y = m|X) = \frac{e^{Z_m}}{\sum_{\ell=0}^{9} e^{Z_\ell}},$$

$$e^1 + e^3 + e^{-1} + e^2 + e^5$$
$$= 178.97$$

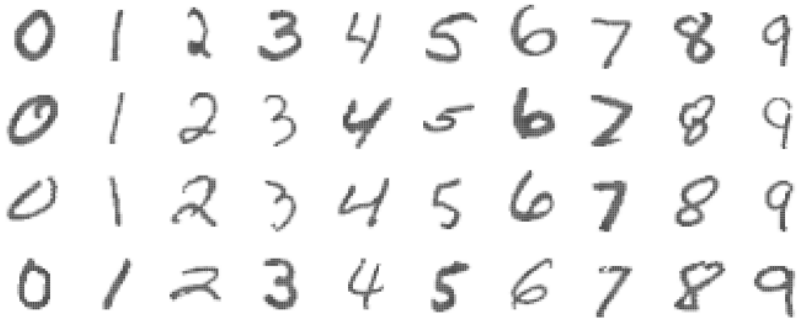
$$Z = \begin{pmatrix} 1 & 3 & -1 & 2 & 5 \end{pmatrix}$$

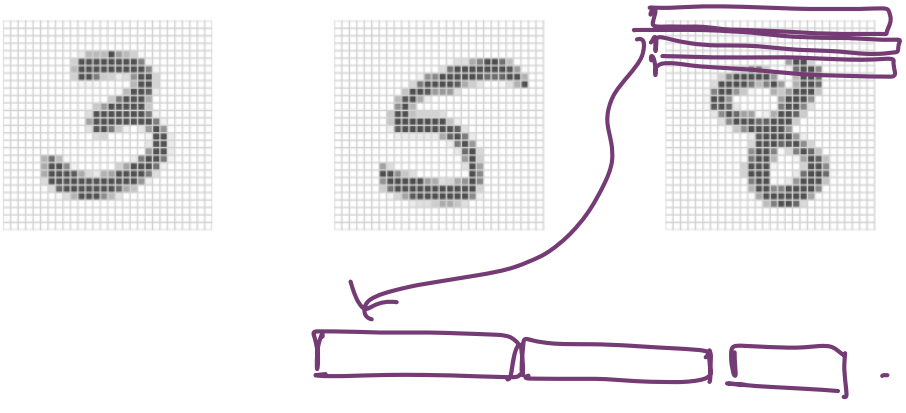
$$f = \left(\frac{e^1}{178.97}, \frac{e^3}{178.97}, \frac{e^{-1}}{178.97}, \ldots \right)$$

$$= (0.01518, 0.1122, 0.00205, 0.04128, 0.82924)$$

this class will be predicted

MNIST





60K training images, 20K test

each image has $p = 28 \times 28 = 784$ pixels

each pixel grayscale value in $[0, 255)$

Goal: predict what number is written

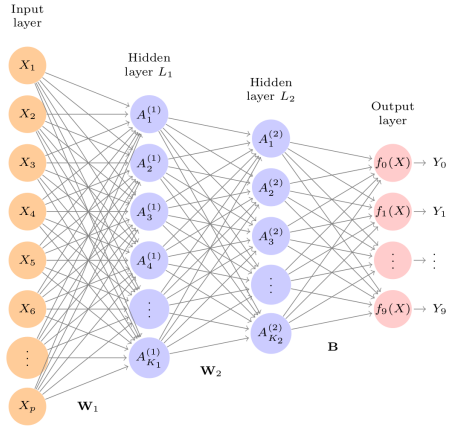
Today: Flatten each picture into a vector length 784

Softmax output to predict #

Coding

TL;DR

Feed Forward Neural Net



$$A_k = h_k(X) = g(w_{k0} + \sum_{j=1}^{p} w_{kj} X_j),$$

- Combines input data using learned weights
- Linear combo of those to get output
- Sometimes softmax to get probability of classification

Next time

Lec #	Date			Reading	HW
21	Mon	10/28	Polynomial & Step Functions	7.1,7.2	
22	Wed	10/30	Step Functions; Basis functions; Start Splines	7.2 - 7.4	
23	Fri	11/1	Regression Splines	7.4	HW #6 Due Sun 11/3
24	Mon	11/4	Decision Trees	8.1	
25	Wed	11/6	Class Cancelled (Dr Munch out of town)		
26	Fri	11/8	Random Forests	8.2.1, 8.2.2	HW #7 Due Sun 11/10
27	Mon	11/11	Maximal Margin Classifier	9.1	
28	Wed	11/13	SVC	9.2	
29	Fri	11/15	SVM	9.3, 9.4	HW #8 Due Sun 11/17
30	Mon	11/18	Single layer NN	10.1	
31	Wed	11/20	Multi Layer NN	10.2	
32	Fri	11/22	CNN	10.3	HW #11 Due Sun 11/24
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